ThinkHub | PROJECT SHOWCASE SERIES

PROJECT 004

CrashLens

ROAD ACCIDENT HOTSPOT DETECTION

"Finding where accidents cluster, not just where they happen."

- Accident Hotspots
- Clustering (KMeans & DBSCAN)
- Severity Analysis
- Geospatial Visualization



By Daksh Khandelwal
ThinkHub

ThinkHub | CrashLens

PROJECT 004

THE PROBLEM

Road Accidents: A Persistent Real-World Challenge



A Serious Issue

Road accidents cause significant loss of lives, injuries and economic cost worldwide.

Why Single Records Aren't Enough?

Individual accident records tell us where an accident happened, but not where **accidents concentrate or repeat.**

Why Hotspots Matter?

Identifying geographical hotspots helps authorities prioritize high-risk areas for better planning, signage, and enforcement.

Individual Records



Shows accidents, but **no clear pattern.**

Accident Hotspots



Reveals high-risk areas where accidents cluster.

The Need: *Move from isolated accident points to data-driven **hotspot intelligence**.*

ThinkHub | CrashLens

PROJECT 004

THE SOLUTION

CrashLens: From Accident Data to Road Safety Intelligence

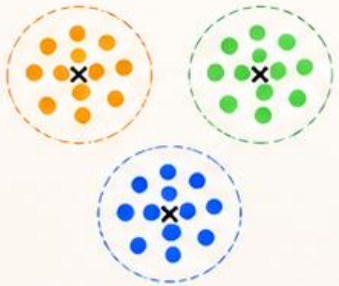
A **geospatial ML platform** that discovers where road accidents cluster, analyzes their severity, and transforms them into **actionable safety insights.**



STATS19 Data → Data Cleaning → Clustering (KMeans & DBSCAN) → Hotspot Ranking → Interactive Visualization

KMeans

Baseline

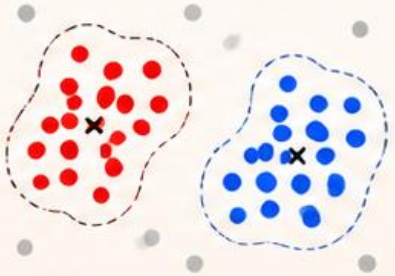


- Centroid-based clustering
- Splits data into K compact clusters
- Evaluated using WCSS (Inertia) & Silhouette Score

VS

DBSCAN

Primary Detector



- Density-based clustering
- Uses Haversine geographical distance
- Finds arbitrarily shaped hotspots
- Identifies noise / isolated points

Why Geographical Density Matters?

- Reveals accident-prone zones (e.g., intersections, roundabouts, corridors)
- Helps prioritize road safety interventions
- More useful than just plotting individual accident locations

In Short: *CrashLens uses KMeans for a baseline view and DBSCAN as the primary method to uncover real accident hotspots through spatial density.*

ThinkHub | CrashLens

PROJECT 004

THE TECHNOLOGY

Technology Behind CrashLens

A modern stack that combines **real-world data**, **unsupervised machine learning**, and **interactive geospatial visualization**.



Data Source

UK Department for Transport | STATS19 Road Casualty Statistics

- Official UK road accident dataset
- Greater London (26,030 records)
- Open Government Licence (OGL v3.0)

Machine Learning

python | scikit-learn

- **KMeans:** Centroid-based clustering (baseline comparison)
- **DBSCAN:** Density-based clustering (primary hotspot detector)
- **Haversine Distance:** Geographical distance on Earth's surface (R = 6,371 km)

Web Application

NEXT.JS | React | TypeScript | Tailwind CSS | Leaflet | Recharts

- Interactive map and data exploration
- Modern, responsive user interface
- Built with Next.js, TypeScript and Leaflet

Tech Stack at a Glance

Python | Pandas | Scikit-learn | Next.js | React | TypeScript | Leaflet | Recharts



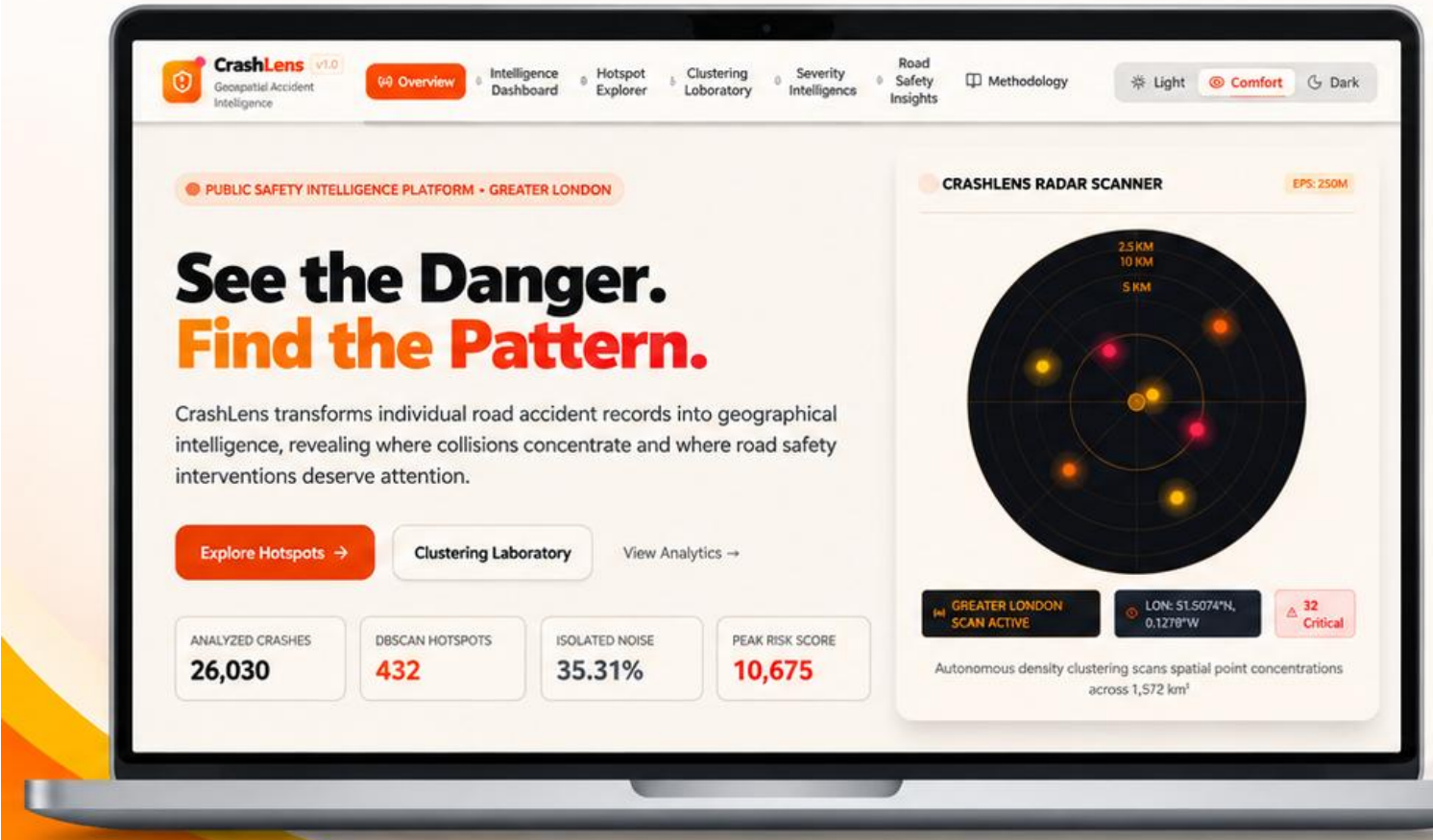
ThinkHub | CrashLens

PROJECT 004

THE PLATFORM

Introducing the CrashLens Platform

A complete, interactive web platform for accident hotspot detection and road safety intelligence.



Interactive Hotspot Map

ML-Based Clustering

Temporal & Severity Insights

Live Website

https://crashlens-snowy.vercel.app/

Let's Explore the Platform Live!